Quantum Field Theory III Week 7 Homework

Exercise 7.5

O(N)-symmetric scalar field theory, linear sigma model:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}\mu^2\phi^2 - \frac{\lambda}{4}(\phi^2)^2 + \frac{1}{2}\delta_Z(\partial_\mu\phi)^2 - \frac{1}{2}\delta_\mu\phi^2 + \frac{1}{4}\delta_\lambda(\phi^2)^2$$

where $\phi^2 = \phi^a\phi^a$, $a = 1, \dots, N$. We assume spontanious symmetry braking, $\mu^2 < 0$. Let's expand around the vacuum. Without loss of generality we can take the vacuum to lie in N-th direction

$\phi_0^i = (0, 0, \cdots, 0, v)$ is the vacuum now let's expand around this with the shift:

$$\phi^i(x) = (\pi^1, \pi^2, \dots, \pi^{N-1}, \sigma) + \phi_0^i$$

Let's write this as

$$\phi^i(x) = (\pi^k, v + \sigma)$$

with $\underline{\underline{v = +\frac{|\mu|}{\sqrt{\lambda}}}}$. Now lets write the Lagrangian in terms of π and σ

Let's start by writing $(\phi^i)^2$: $(\phi^i)^2 = \phi^a\phi^a = (\pi^k)^2 + (v+\sigma)^2$.

Then we get $((\phi^i)^2)^2 = (\pi^k)^4 + 2(\pi^k)^2(v+\sigma)^2 + (v+\sigma)^4$

$$\Rightarrow (\phi^2)^2 = (\pi^k)^4 + 2(\pi^k)^2v^2 + 4(\pi^k)^2v\sigma + 2(\pi^k)^2\sigma^2 + v^4 + 4v^3\sigma + 4v\sigma^3 + 6v^2\sigma^2 + \sigma^4$$

We can always drop constant terms from Lagrangian since it doesn't change the calculation: Let's drop v^4 term ($(\pi^k)^4 \equiv ((\pi^k)^2)^2$ here and forward)

$$\Rightarrow (\phi^2)^2 = (\pi^k)^4 + 2(\pi^k)^2v^2 + 4v\sigma(\pi^k)^2 + 2(\pi^k)^2\sigma^2 + 4v^3\sigma + 4v\sigma^3 + 6v^2\sigma^2 + \sigma^4$$

And the other terms in Lagrangian become:

$(\partial_\mu\phi^i)^2 = (\partial_\mu\pi^k)^2 + (\partial_\mu\sigma)^2$, since $\partial_\mu v = 0$ since v is constant

$\phi^2 = (\pi^k)^2 + (v+\sigma)^2 = (\pi^k)^2 + 2\sigma v + \sigma^2$, where we dropped v^2 term for being a constant.

Now our Lagrangian is:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\pi^k)^2 + \frac{1}{2}(\partial_\mu\sigma)^2 - \frac{1}{2}\mu^2(\pi^k)^2 - \mu^2\sigma v - \frac{1}{2}\mu^2\sigma^2 - \frac{\lambda}{4}\Big[(\pi^k)^4 + 2(\pi^k)^2v^2 + 4v(\pi^k)^2\sigma + 2(\pi^k)^2\sigma^2$$
$$+ 4v^3\sigma + 4v\sigma^3 + 6v^2\sigma^2 + \sigma^4\Big] + \frac{1}{2}\delta_Z(\partial_\mu\pi^k)^2 + \frac{1}{2}\delta_Z(\partial_\mu\sigma)^2 - \frac{1}{2}\delta_\mu(\pi^k)^2 - v\delta_\mu\sigma - \frac{1}{2}\delta_\mu\sigma^2$$
$$+ \frac{1}{4}\delta_\lambda\Big[(\pi^k)^4 + 2(\pi^k)^2v^2 + 4v(\pi^k)^2\sigma + 2(\pi^k)^2\sigma^2 + 4v^3\sigma + 4v\sigma^3 + 6v^2\sigma^2 + \sigma^4\Big]$$

We see that $-\frac{1}{2}\mu^2(\pi^k)^2 - \frac{\lambda}{2}(\pi^k)^2v^2 = 0$ since $\mu^2 < 0$ and $v^2 = \frac{|\mu|^2}{\lambda}$

let's also list some further simplifications:

$$-\frac{1}{2}\mu^2\sigma^2 - \frac{\lambda}{4}\left[6v^2\sigma^2\right] = -\frac{1}{2}\mu^2\sigma^2 - \frac{1}{2}\left(3|\mu|^2\sigma^2\right) = -\frac{1}{2}\left(2|\mu|^2\right)\sigma^2$$

$$-\mu^2\sigma v - \frac{\lambda}{4}\left(4v^3\sigma\right) = -\mu^2\sigma v - |\mu|^2 v\sigma = 0 \qquad \text{since} \quad \mu^2 = -|\mu|^2$$

Using These simplification our Lagrangian is now -

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\pi^k)^2 + \frac{1}{2}(\partial_\mu\sigma)^2 - \frac{1}{2}\left(2|\mu|^2\right)\sigma^2 - \lambda v\sigma^3 - \lambda v(\pi^k)^2\sigma - \frac{\lambda}{4}\sigma^4 - \frac{\lambda}{2}(\pi^k)^2\sigma^2$$

$$- \frac{\lambda}{4}(\pi^k)^4 + \frac{1}{2}\delta_Z(\partial_\mu\pi^k)^2 + \frac{1}{2}\delta_Z(\partial_\mu\sigma)^2 + (\pi^k)^2\left(-\frac{1}{2}\delta_\mu + \frac{\delta_\lambda}{4}\left(2v^2 + 4v\sigma + 2\sigma^2\right)\right)$$

$$+ \sigma\left(-v\delta_\mu + \frac{\delta_\lambda}{4}\left(4v^3\right)\right) + \sigma^2\left(-\frac{1}{2}\delta_\mu + \frac{\delta_\lambda}{4}\left(6v^2\right)\right) + \frac{\delta_\lambda}{4}(\pi^k)^4 + \frac{\delta_\lambda}{4}\sigma^4$$

$$+ \delta_\lambda v\sigma^3$$

$$\Rightarrow \mathcal{L} = \frac{1}{2}(\partial_\mu\pi^k)^2 + \frac{1}{2}(\partial_\mu\sigma)^2 - \frac{1}{2}\left(2|\mu|^2\right)\sigma^2 - \lambda v\sigma^3 - \lambda v(\pi^k)^2\sigma - \frac{\lambda}{4}\sigma^4 - \frac{\lambda}{2}(\pi^k)^2\sigma^2 - \frac{\lambda}{4}(\pi^k)^4$$

$$+ \frac{1}{2}\delta_Z(\partial_\mu\pi^k)^2 - \frac{1}{2}\left(\delta_\mu - \delta_\lambda v^2\right)(\pi^k)^2 + \frac{1}{2}\delta_Z\left(\partial_\mu\sigma^2\right) - \frac{1}{2}\left(\delta_\mu - 3\delta_\lambda v^2\right)\sigma^2 + \frac{\delta_\lambda}{4}\sigma^4$$

$$- \left(\delta_\mu v + \delta_\lambda v^3\right)\sigma + \delta_\lambda v\sigma(\pi^k)^2 + \delta_\lambda v\sigma^3 + \frac{\delta_\lambda}{4}(\pi^k)^4 + \frac{\delta_\lambda}{2}\sigma^2(\pi^k)^2$$

Now we almost have the Lagrangian in the desired form.
To get the correct Feynman rules, that can be direcltly used for calculation without having to thik about combinatoric factors that come from contracting the fields, we need to account for them in the Lagrangian:

σ^3 is basically a ϕ^3-theory 3-point interacton so a factor of $\frac{1}{3!}$ is helpful to calculate the Feynman rule:

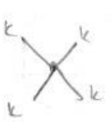
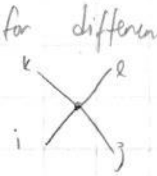
$\lambda v\sigma^3 \to \frac{1}{3!}(6\lambda v\sigma^3)$ which given Feynmann rule of  $= -6i\lambda v$

Similarily to ϕ^4-theory we have factor of $\frac{1}{4!}$ for σ^4 so we get

$\frac{\lambda}{4}\sigma^4 \to \frac{1}{4!}(6\lambda\sigma^4)$, which gives us  $= -i6\lambda$

For π^k fields we need to also take into accond different indices/fields

So $\frac{\lambda}{4}(\pi^k)^4 \to \frac{\lambda}{4!}(6(\pi^k)^4)$ which for only a single field gives

 $= -6i\lambda$ but for different field such that the diagram would be  we need to consider more

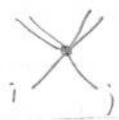
the fields π^k's leaving the vertex and coming to it must be same since a field is it's own anti-particle for scalar fields so we need terms like

δ^{ik} and δ^{lj}, there is 3 ways to do the kronecker deltas so we add symmetry factor of $\frac{1}{3}$ and get

$$-6i\lambda\left[\tfrac{1}{3}\delta^{lj}\delta^{ik}+\tfrac{1}{3}\delta^{ij}\delta^{kl}+\tfrac{1}{3}\delta^{il}\delta^{jk}\right] = -2i\lambda\left[\delta^{ik}\delta^{lj}+\delta^{ij}\delta^{kl}+\delta^{il}\delta^{ik}\right]$$

and this gives correct rules for $i=j=k=l$ which is $-6i\lambda$.

Now for interactions between π^k and σ-field we have two vertex of the form:

 $\sim \lambda\delta^{ij}$ and  $\sim \lambda v\,\delta^{ij}$

Now we need to find the number that multiplies them.

While I have some speculations, I don't really have good explanation for these two vertices so I will just list constants to multiply them by:

$$\lambda v(\pi^k)^2\sigma \to \tfrac{1}{2}\left(2\lambda v(\pi^k)^2\sigma\right) \qquad \text{and} \qquad \tfrac{\lambda}{2}(\pi^k)^2\sigma^2 \to \tfrac{1}{4}\left(2\lambda(\pi^k)^2\sigma^2\right)$$

Which gives us the Feynman rules:

 $= -2i\lambda v\,\delta^{ij}$ and  $= -2i\lambda\,\delta^{ij}$

To get correct constants for counter-terms we just need to match them with ones for non-counter-term counterparts eg if we have $\frac{1}{2}(2\lambda v\sigma(\pi^k)^2)$ then we do $\delta_\lambda v\sigma(\pi^k)^2 = \frac{1}{2}(2\delta_\lambda v\sigma(\pi^k)^2)$ So our Lagrangian now looks like

$$\begin{aligned}\mathcal{L} = {} & \tfrac{1}{2}(\partial_\mu\pi^k)^2 + \tfrac{1}{2}(\partial_\mu\sigma)^2 - \tfrac{1}{2}(2|\mu|^2)\sigma^2 - \tfrac{1}{3!}(6\lambda v)\sigma^3 - \tfrac{1}{4!}(6\lambda)\sigma^4 - \tfrac{1}{4!}(6\lambda)(\pi^k)^4 \\ & - \tfrac{1}{2}(2\lambda v)\sigma(\pi^k)^2 - \tfrac{1}{4}(2\lambda)\sigma^2(\pi^k)^2 + \tfrac{1}{2}\delta_Z(\partial_\mu\pi^k)^2 - \tfrac{1}{2}(\delta_\mu - \delta_\lambda v^2)(\pi^k)^2 - \tfrac{1}{4!}(-6\delta_\lambda)(\pi^k)^4 \\ & + \tfrac{1}{2}\delta_Z(\partial_\mu\sigma^2) - \tfrac{1}{2}(\delta_\mu - 3\delta_\lambda v^2)\sigma^2 + \tfrac{1}{4!}(6\delta_\lambda)\sigma^4 - (\delta_\mu v - \delta_\lambda v^3)\sigma \\ & - \tfrac{1}{2}(-2\delta_\lambda v)\sigma(\pi^k)^2 - \tfrac{1}{3!}(-6\delta_\lambda v)\sigma^3 - \tfrac{1}{4}(-2\delta_\lambda)\sigma^2(\pi^k)^2\end{aligned}$$

The Lagrangian is now in the form where we can directly read the Feynman-rules from it, since the normalization is done such that λ-in every Feynman-rule corresponds to same value of lambda (so no need to do multiplications or divisions to get different lambdas to align for example from experiments.)

Now by directly reading the Feynman rules from the Lagrangian we get:

σ  $= \dfrac{i}{p^2 - 2|\mu|^2}$, π^i  $\pi^j = \dfrac{i\delta^{ij}}{p^2}$

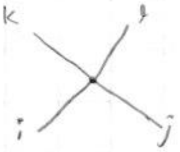
no mass term in Lagrangian

 $= -6i\lambda v$

$= -2i\delta^{ij}\lambda v$

 $= -6i\lambda$ ← derived earlier → 

$= -2i\lambda\delta^{ij}$

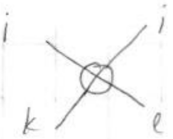


$= -2i\lambda\left[\delta^{ij}\delta^{kl} + \delta^{ik}\delta^{lj} + \delta^{il}\delta^{kj}\right]$

And counter terms can can also be read from the Lagrangian:

 $= i(\delta_Z p^2 - \delta_\mu + 3\delta_\lambda v^2)$

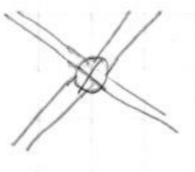
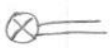
$= i(\delta_Z p^2 - \delta_\mu + \delta_\lambda v^2)\delta^{ij}$

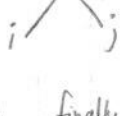


$= +2i\delta_\lambda\left[\delta^{ij}\delta^{kl} + \delta^{ik}\delta^{jl} + \delta^{il}\delta^{jk}\right]$

for same reason as for non-counter term

The relative − sign of δ_λ compared to Peskin comes from our original Lagrangian having $+\delta_\lambda(\cdots)$ as counter term instead of $-\delta_\lambda(\cdots)$ in Peskin

 $= +6i\delta_\lambda$,  $= -i(\delta_\mu v - \delta_\lambda v^3)$



$= +2i\delta^{ij}\delta_\lambda v$,  $= +6i\delta_\lambda v$

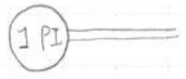
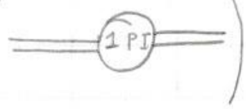
and finally

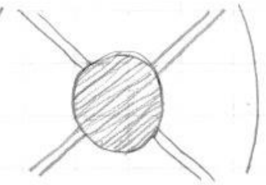
$= +2i\delta_\lambda\delta^{ij}$,

These are the Feynman rules along with the counter terms.

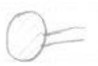
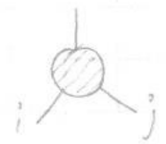
Exercise 7.6

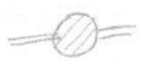
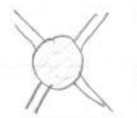
The renormalization conditions are:

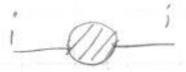
 $= 0$, $\dfrac{d}{dp^2}\Bigg($  $\Bigg) = 0$ at $p^2 = m^2$

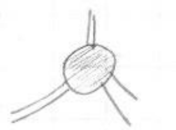
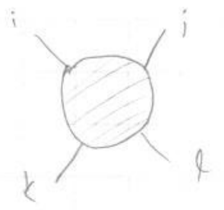
$\mathrm{Im}\Bigg($  $\Bigg) = -6i\lambda$ at $s = 4m^2$, $t = u = 0$

The divergent amplitudes are (From Peskin Figure 11.4), with their degree of divergence listed.

 $D = 3$  $D = 1$

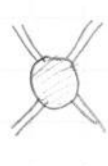
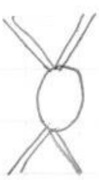
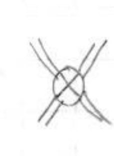
 $D = 2$  $D = 0$,  $D = 0$

 $D = 2$

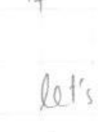
 $D = 1$  $D = 0$

Let's start by calculating divergence of  to fix δ_λ (using Renormalization condition)
At one loop order we have

terms related to crossing symmetry of diagrams so t-, and u-channel diagrams

 $=$  $+$  $+$ (crossing terms) $+$ Finite terms (in UV) $+$ 

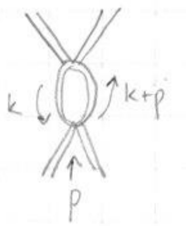
The finite loops are finite since they have three or more propagators which makes the integral finite in UV; for example

 $\sim \displaystyle\int \frac{d^4k}{(2\pi)^4}\frac{1}{k^2}\frac{1}{k^2}\frac{1}{k^2} = \int d\Omega \int \frac{dk}{(2\pi)^4}\frac{k^3}{k^6} = \int d\Omega \int \frac{dk}{(2\pi)^4}\frac{1}{k^3} = \text{finite}$

here we neglect external legs momentum $\Rightarrow$ assume small

while if there is 2 propagators we have $\displaystyle\sim\int\frac{d^4k}{(2\pi)^4}\frac{1}{k^2}\frac{1}{k^2} = \int d\Omega\int\frac{dk}{(2\pi)^4}\frac{1}{k} \sim \log\text{-divergence}$

So let's now calculate the divergent part of the divergent loops, since we are only interested in the divergences of different diagrams.



4-point-vertex

✓ We use dimensional regularization.

$$= \underbrace{\frac{1}{2}}_{\text{Symmetry factor}} \cdot (-6i\lambda)^2 \cdot \int \frac{d^dk}{(2\pi)^d} \frac{i}{k^2 - 2|\mu|^2} \frac{i}{(k+p)^2 - 2|\mu|^2}$$

propagators

Let's use Feynman parametrization:

the denominator becomes $\left[(1-x)k^2 - (1-x)2|\mu|^2 + x(k^2+2kp+p^2) - 2x|\mu|^2\right]^2$

$$= \left[+k^2 + 2xkp + xp^2 - 2|\mu|^2\right]^2 = \left[\ell^2 - \Delta\right]^2 \quad \text{with } \ell = (k + xp)$$

and Δ is function of p and $|\mu|^2$ that we don't need to think about since it doesn't appear in divergent part

$\Rightarrow$ the loop is now, with shift of $\ell = k + xp$, $dk = d\ell$

$$\text{loop} = 18\lambda^2 \int dx \int \frac{d^d\ell}{(2\pi)^d} \frac{1}{(\ell^2 - \Delta)^2}$$

Now let's directly use the d-dimensional Minkowski space integral results from Peskin (A.44), which gives us

$$\text{loop} = 18\lambda^2 \int dx \frac{i}{(4\pi)^{d/2}} \frac{\Gamma(2-d/2)}{\Gamma(2)} \left(\frac{1}{\Delta}\right)^{2-d/2}$$

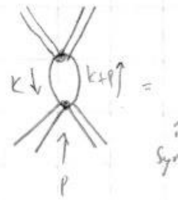
which we can split into finite and divergent parts

$$\text{loop} = i\,18\lambda^2 \frac{\Gamma(2-d/2)}{(4\pi)^2} + (\text{finite terms})$$

$\left(\frac{1}{\Delta}\right)^{2-d/2} = 1$ for $d=2$ and then there is no x-dependence for divergent part

So for diagram  we get the divergence to be $\dfrac{18i\lambda^2\,\Gamma(2-d/2)}{(4\pi)^2}$

Now let's calculate the divergence of 



$$= \underbrace{\frac{1}{2}}_{\text{Symmetry}} (N-1) \underbrace{(-2i\lambda)^2}_{\text{vertex factor}} \int \frac{d^dk}{(2\pi)^d} \frac{i}{k^2} \frac{i}{(k+p)^2}, \quad \pi^k \text{ is massless}$$

there are $(N-1)$ possible π^k fields, so $(N-1)$ factors for i,j in $-2i\lambda\delta^{ij}$ of vertex factor

$$\Rightarrow \text{loop} = 2\lambda^2(N-1) \int \frac{d^dk}{(2\pi)^d} \int dx \frac{1}{(1-x)k^2 + x(k^2+2kp+p^2)}$$

Feynman parametrization use $\ell = k + xp$ and Δ which is function of p

$$= 2\lambda^2(N-1) \int dx \int \frac{d^d\ell}{(2\pi)^d} \frac{1}{[\ell^2 - \Delta]^2}$$

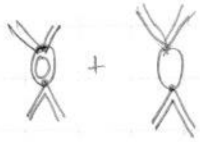
, Now use integral result from (A.44) Peskin

$$= 2\lambda^2(N-1) \int dx \frac{i}{(4\pi)^{d/2}} \Gamma(2-d/2) \left(\frac{1}{\Delta}\right)^{2-d/2}$$

, which we can split to finite and divergent parts:

$$= 2i\lambda^2(N-1) \frac{\Gamma(2-d/2)}{(4\pi)^2} + (\text{finite terms})$$

We see that the divergent parts are independent of momentum so we get Same divergence for their crossing counterparts (t, u, channel)
So the total divergence is ~~is~~ 3 times the sum of two divergences we got:

$\Rightarrow$  + (crossing) $\overset{\text{divergent part only}}{=}$ $3\cdot\left[18i\lambda^2 + 2i\lambda^2(N-1)\right]\frac{\Gamma(2-d/2)}{(4\pi)^2}$

$$= 6i\lambda^2\left[N+8\right]\frac{\Gamma(2-d/2)}{(4\pi)^2}$$

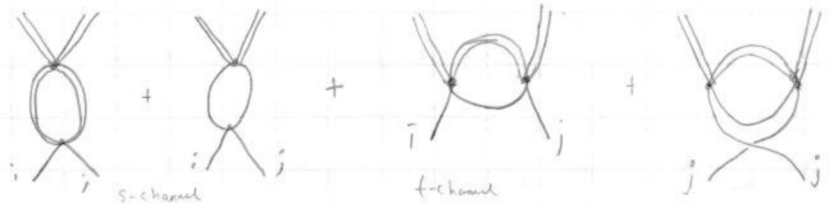
The Feynman rule for  was $+6i\delta_\lambda$ which means that the counter term δ_λ needs to be $-\lambda^2(N+8)\,\Gamma(2-d/2)\,(4\pi)^{-2}$ to cancel the divergence

$$\boxed{\Rightarrow \quad \delta_\lambda = -\lambda^2(N+8)\frac{\Gamma(2-d/2)}{(4\pi)^2}}$$

Relative minus sign compared to Peskin again comes from different convention used in Lagrangian.

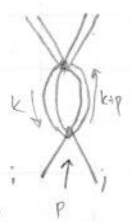
We have gotten the counter-term from the Renormalization condition let's see if this is enough to cancle divergencies of other diagrams that have δ_λ in their counter terms

Let's start with , just like in previous σ^4 diagram the loops with 3 or more propagators are finite, which leaves us with following divergent loops:



[Note: I have been using peskin notation for diagram where time runs vertically]

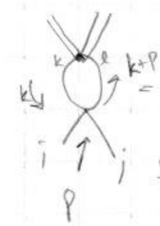
Now we can start calculating divergent parts of these integrals

 $= \underset{\text{Symmetry factor}}{\frac{1}{2}}\ \underbrace{(-6i\lambda)}_{\text{4-point vertex of }\sigma^4}\ \underbrace{(-2i\lambda\delta^{ij})}_{(\pi^i)^2\sigma^2\text{-vertex}}\ \underbrace{\int\frac{d^dk}{(2\pi)^d}\frac{i}{k^2-2|\mu|^2}\frac{i}{(k+p)^2-2|\mu|^2}}$

This loop integral we have already done for , so lets use the result:

 $= 6\lambda^2\delta^{ij}\cdot\int dx\,\frac{i}{(4\pi)^{d/2}}\frac{\Gamma(2-\frac{d}{2})}{\Gamma(2)}\left(\frac{1}{\Delta}\right)^{2-d/2} = 6i\lambda^2\delta^{ij}\frac{\Gamma(2-\frac{d}{2})}{(4\pi)^2} + (\text{finite})$

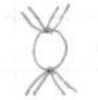
Now let's move on to the next loop integral



vertex terms

$$= \underbrace{\frac{1}{2}}_{\text{Symmetry factor}} (-2i\lambda\delta^{kl})\left(-2i\lambda(\delta^{ij}\delta^{kl}+\delta^{ik}\delta^{jl}+\delta^{il}\delta^{jk})\right) \underbrace{\int \frac{d^dk}{(2\pi)^d}\frac{i}{k^2}\frac{i}{(k+p)^2}}$$

$$\text{loop} = 2\lambda^2\delta^{kl}(\delta^{ij}\delta^{kl}+\delta^{ik}\delta^{jl}+\delta^{il}\delta^{jk})\int\frac{d^dk}{(2\pi)^d}\frac{1}{k^2(k+p)^2}$$

This is same integral as the one done for  - integral

Let's use the result for the integral

$$= 2\lambda^2(N+1)\delta^{ij}\left[i\frac{\Gamma(2-d/2)}{(4\pi)^2} + (\text{finite terms})\right]$$

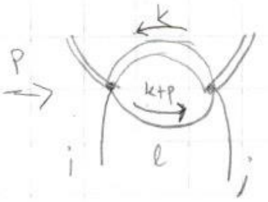
where the $N+1$ terms comes from

$$\delta^{kl}(\delta^{ij}\delta^{kl}+\delta^{ik}\delta^{jl}+\delta^{il}\delta^{jk})$$
$$= \delta^{ij}(N-1)+\delta^{ik}\delta^{jk}+\delta^{ik}\delta^{jk}$$
$$= \delta^{ij}(N-1)+\delta^{ij}+\delta^{ij} = \delta^{ij}(N+1)$$

So the divergent part of the diagram is:

$$2i\lambda^2(N+1)\delta^{ij}\frac{\Gamma(2-d/2)}{(4\pi)^2}$$

Now let's evaluate the diagram



No symmetry factors

$$= 1\cdot(-2i\lambda\delta^{il})(-2i\lambda\delta^{lj})\int\frac{d^dk}{(2\pi)^d}\frac{i}{k^2+2|\mu|^2}\frac{i}{(k+p)^2}$$

We can again with Feynman parameters and some shift write the integral as

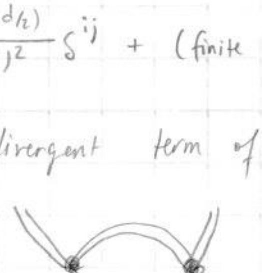
$$\int\frac{d^dk}{(2\pi)^d}\frac{1}{k^2+2|\mu|^2}\frac{1}{(k+p)^2} = \int dx\int\frac{d^dl}{(2\pi)^d}\frac{1}{(l^2-\Delta)^2} = \int dx\frac{i}{(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\left(\frac{1}{\Delta}\right)^{2-d/2}$$

$$= \frac{i\Gamma(2-d/2)}{(4\pi)^2} + (\text{finite terms})$$

So we don't need to worry about exact form of Δ to get divergent part

$$\Rightarrow \text{loop} = 4i\lambda^2\frac{\Gamma(2-d/2)}{(4\pi)^2}\delta^{ij} + (\text{finite terms})$$

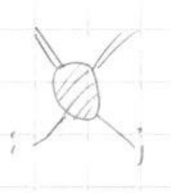
So the loop has divergent term of $4i\lambda^2\delta^{ij}\frac{\Gamma(2-d/2)}{(4\pi)^2}$

The final loop 

is same as previous one with i and j interchanged at least for the divergent term which is the only one we are interested in

$\Rightarrow$ this loop also has same divergence as previous one which is

$$4i\lambda^2\delta^{ij}\frac{\Gamma(2-d/2)}{(4\pi)^2}$$

Now let's add them together to get the 1-loop divergence of 

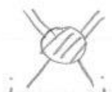
, not including the counter term

$$\Gamma(\text{loops}) \overset{\text{divergence part only of sum of loops}}{=} \left[6i\lambda^2\delta^{ij} + 2i\lambda^2\delta^{ij}(N+1) + 2\left(4i\lambda^2\delta^{ij}\right)\right]\frac{\Gamma(2-d/2)}{(4\pi)^2}$$

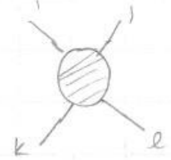
$$= 2i\lambda^2\delta^{ij}(N+8)\frac{\Gamma(2-d/2)}{(4\pi)^2}$$

the counter-term for $\sigma\pi^k\pi^k$ is  $= 2i\,\delta\lambda\,\delta^{ij}$ and

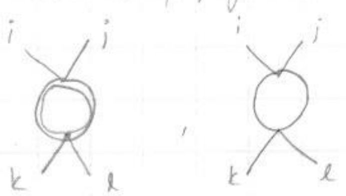
$\delta\lambda = -(N+8)\lambda^2\frac{\Gamma(2-d/2)}{(4\pi)^2}$ indeed cancels the divergence from the 1-loops.

So the counterterm $\delta\lambda$ indeed leads to cancellation of all divergences in 

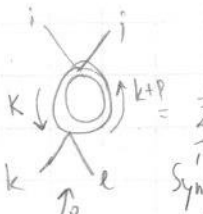
at least up to 1-loop order.

Now let's move on to the diagram 

The only divergent loops for this (ones with less than 3 propagators) are



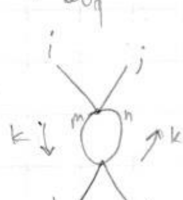
+ (crossing terms), Let's now calculate their divergent terms:



$$= \underbrace{\frac{1}{2}}_{\text{Symmetry factor}}\overbrace{(-2i\lambda\delta^{ij})(-2i\lambda\delta^{kl})}^{\text{vertex factors}}\underbrace{\int\frac{d^dk}{(2\pi)^d}\frac{i}{k^2-2|\mu|^2}\frac{i}{(k+p)^2-2|\mu|^2}}_{}$$

We have already calculated this integral that corresponds to  loop, let's use the resut:

$$\Rightarrow \text{loop} = 2i\lambda^2\delta^{ij}\delta^{kl}\frac{\Gamma(2-d/2)}{(4\pi)^2} + (\text{finite terms})$$, Let's move on to next diagram

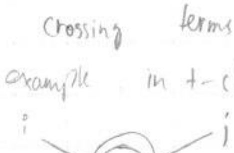


$$= \frac{1}{2}\left(-2i\lambda\left[\delta^{ij}\delta^{mn}+\delta^{im}\delta^{jn}+\delta^{in}\delta^{jm}\right]\right)\left(-2i\lambda\left[\delta^{kl}\delta^{mn}+\delta^{km}\delta^{ln}+\delta^{kn}\delta^{lm}\right]\right)\int\frac{d^dk}{(2\pi)^D}\frac{1}{k^2}\frac{1}{(k+p)^2}$$

$$= 2\lambda^2\left[\delta^{ij}\delta^{kl}(N-1)+4\delta^{ij}\delta^{kl}+2\delta^{ik}\delta^{jl}+2\delta^{il}\delta^{jk}\right]\left(i\frac{\Gamma(2-d/2)}{(4\pi)^2}+(\text{finite})\right)$$ already computed O-loops before

$$= 2i\lambda^2\left[(N+3)\delta^{ij}\delta^{kl}+2\delta^{ik}\delta^{jl}+2\delta^{il}\delta^{jk}\right]\frac{\Gamma(2-d/2)}{(4\pi)^2} + (\text{finite terms})$$

The crossing terms give basically the same diagrams but with different pairs of indices for example in t-channel we have

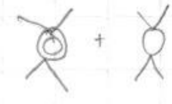


which amounts to change of index in result such that j and k are interchanged compared to our s-channel result for u-channel we get  So now we have interchange of j and l

So the divergent part is $2i\lambda^2 \delta^{il}\delta^{jk} \frac{\Gamma(2-d/2)}{(4\pi)^2}$ for u-channel and $2i\lambda^2 \delta^{ik}\delta^{jl} \frac{\Gamma(2-d/2)}{(4\pi)^2}$ for t-channel when the loop is σ-loop.
For Π^k loop the same principle applies $j \leftrightarrow k$ for t-channel and $j \leftrightarrow l$ for u-channel so that the divergent parts are :

t-channel : $2i\lambda^2\left((N+3)\delta^{ik}\delta^{jl} + 2\delta^{ij}\delta^{kl} + 2\delta^{il}\delta^{jk}\right)\frac{\Gamma(2-d/2)}{(4\pi)^2}$

u-channel : $2i\lambda^2\left[(N+3)\delta^{il}\delta^{jk} + 2\delta^{ik}\delta^{jl} + 2\delta^{ij}\delta^{kl}\right)\frac{\Gamma(2-d/2)}{(4\pi)^2}$

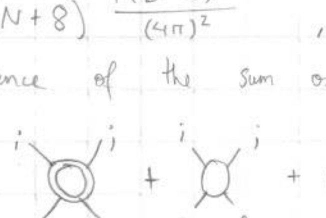
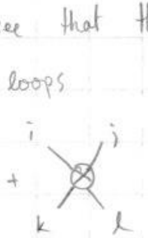
Summing these together we get divergent part of  + (crossing terms) to be

$$2i\lambda^2\left[\delta^{ij}\delta^{kl} + \delta^{ik}\delta^{jl} + \delta^{il}\delta^{jk}\right]\Big(\underbrace{(N+3)}_{\text{From } \Pi^k \text{ loop}} + \underbrace{1}_{\text{from } \sigma\text{-loop}} + 2 + 2\Big)\frac{\Gamma(2-d/2)}{(4\pi)^2}$$

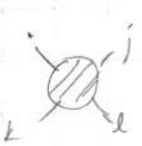
$$= 2i\lambda^2\left[\delta^{ij}\delta^{kl} + \delta^{ik}\delta^{jl} + \delta^{il}\delta^{jk}\right](N+8)\frac{\Gamma(2-d/2)}{(4\pi)^2}$$

The counter term is  $= +2i\delta_\lambda\left[\delta^{ij}\delta^{kl} + \delta^{ik}\delta^{jl} + \delta^{il}\delta^{jk}\right]$

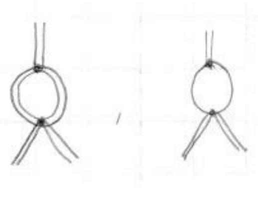
with $\delta_\lambda = -\lambda^2(N+8)\frac{\Gamma(2-d/2)}{(4\pi)^2}$, we can see that the counter term indeed cancels the divergence of the sum of divergent loops

$\Rightarrow$  =  + (crossing terms) +  + Finite terms

$= 0$

 = Finite, to 1-loop order

Now let's look into , since the loops with 3-or-more propagators are finite we only have following diagrams

 and (crossing diagrams)

Let's start with  $= \frac{1}{2}\cdot(-6i\lambda)\underbrace{(-6i\lambda v)}_{\text{Four point vertex}}\int\frac{d^dk}{(2\pi)^d}\frac{i}{k^2-2|\mu|^2}\frac{i}{(k+p)^2-2|\mu|^2}$

this is just  but we have v instead of one of the legs so only chang is $(-6i\lambda) \to (-6i\lambda v)$ from going to 3-point function from 4-point function. This holds for the loop

 too since  $= -2i\lambda\delta^{ij}$ and  $= -2i\lambda v\delta^{ij}$

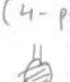
So we can see that this works for crossing terms too and since in it too we replace a 4-point vertex with 3-point vertex. So a factor of v comes for removing a external leg from a vertex (that still remains a vertex afterwards), this happens for all our divergent terms since in  there are only 4-point vertices connected to loop
So we can relate the divergent parts by divergent (3-point) = $v\cdot$ divergent (4-point)
This relation also holds for counter terms.

So the divergent part of  +  + (crossing diagrams) is

$$v\cdot\left(6i\lambda^2\,[N+8]\,\frac{\Gamma(2-d/2)}{(4\pi)^2}\right)$$ meanwhile the counter-term is

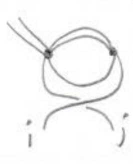
 $= 6i\,\delta_\lambda v$ with $\delta_\lambda = -\lambda^2(N+8)\dfrac{\Gamma(2-d/2)}{(4\pi)^2}$

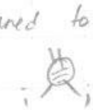
which completely cancels the divergence in  +  + (crossing)

making  finite. This holds becaus the vertex are related via
3-point vertex = $v\cdot$ (4-point vertex) and the amount of propagator in divergent terms of  are same as in divergent part of  (Not counting counter terms in these) making the loop integrals of the diagrams identical.

We can show this reasoning also holds when comparing  and :

Now let's move on to  which has the diverent loops

, ,  and 

These are all same loops as in  but with one of the 4-point $\sigma\sigma\pi\pi$-vertex replaced by $\sigma\pi\pi$ 3-point vertex which are related via
(3-point vertex) = $v\cdot$ (4-point vertex) as we can see from the Feynman rules.
And because the loop integrals are identical here compared to divergent loops of  the divergent terms that arise from  is jus v-times the divergent terms that arise from 
$\Rightarrow$ the divergent terms of  are $v\cdot\left(2i\lambda^2\delta^{ij}(N+8)\dfrac{\Gamma(2-d/2)}{(4\pi)^2}\right)$

the counter-term is  $= 2i\,\delta^{ij}\,\delta_\lambda\,v$ with

$$\delta_\lambda = -\lambda^2 (N+8)\,\Gamma(2-d/2)/(4\pi)^2$$

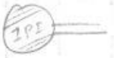
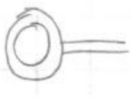
$\Rightarrow$  $= -\left(2i\lambda^2\delta^{ij}(N+8)\dfrac{\Gamma(2-d/2)}{(4\pi)^2}\right)\cdot v$, and we can see that

this totally cancels the divergence of the sum of divergent loops.

$\Rightarrow$  is finite

We have shown that all amplitudes with only δ_λ in counter-term are finite at least up to 1-loop order. Now let's examine rest of the Amplitudes starting with tadpole:

The first renormalization condition gives us

 $= 0 \Rightarrow$  $+$  $+$  $= 0$

Let's use this renormalization condition to get expression for δ_μ

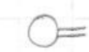
 $= \underbrace{\frac{1}{2}}_{\text{Symmetry factor}}(-6i\lambda v)\displaystyle\int\frac{d^dk}{(2\pi)^d}\frac{i}{k^2-2|\mu|^2}$, We can use integral from Peskin (A.44) directly

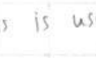
 $= -3i^2\lambda v\,\dfrac{-i}{(4\pi)^{d/2}}\,\dfrac{\Gamma(1-d/2)}{\Gamma(1)}\left(\dfrac{1}{2|\mu|^2}\right)^{1-d/2}$ || $\Gamma(1-d/2)$ is divergent for $d=4$

The second term is divergent with integral over a massless propagator. Let's a small mass parameter ξ for π-field as an infrared regulator

$\Rightarrow$  $= \dfrac{1}{2}(-2i\lambda v\delta^{ij})\displaystyle\int\frac{d^dk}{(2\pi)^d}\frac{i\delta^{ij}}{k^2-\xi^2} = -i^2\lambda v(N-1)\frac{-i}{(4\pi)^{d/2}}\Gamma(1-d/2)\left(\frac{1}{\xi^2}\right)^{1-d/2}$

Now let's use renormalization condition with  $= -i(\delta_\mu v - \delta_\lambda v^3)$

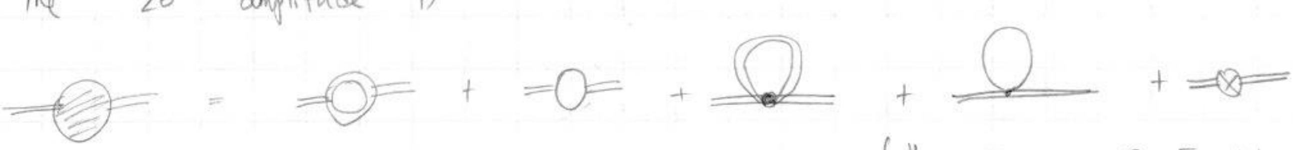
$\Rightarrow$ $i(\delta_\mu - \delta_\lambda v^2)v = -\lambda i v\dfrac{\Gamma(1-d/2)}{(4\pi)^{d/2}}\left[\dfrac{3}{(2|\mu|^2)^{1-d/2}} + \dfrac{N-1}{(\xi)^{1-d/2}}\right] =$  $+$ 

$\Rightarrow$ $(\delta_\mu - \delta_\lambda v^2) = -\lambda\dfrac{\Gamma(1-d/2)}{(4\pi)^{d/2}}\left(\dfrac{3}{(2|\mu|^2)^{1-d/2}} + \dfrac{N-1}{(\xi)^{1-d/2}}\right) = \dfrac{1}{vi}\Big[$  $+$  $\Big]$

this is useful for later

This is the expression for δ_μ we get from first renormalization condition and it is fine to leave it as it is sinse this form will be usefull with two-point amplitude

The 2σ amplitude is



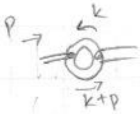
(there is no $\sigma\sigma\pi$ - interaction)

where the counter term vertex is

 $= i\left(\delta_Z p^2 - \delta_\mu + 3\,\delta_\lambda v^2\right)$

$= i\delta_Z p^2 - i\left(\delta_\mu - v^2\delta_\lambda\right) + 2i\,\delta_\lambda v^2$

Let's first calculate the divergent terms of  and 

 $= \frac{1}{2}\left(-6i\lambda v\right)^2 \underbrace{\int \frac{d^dk}{(2\pi)^d}\frac{i}{k^2-2|\mu|^2}\frac{i}{(k+p)^2-2|\mu|^2}}_{\text{We already calculated this, so let's use the result}}$

$= +18\lambda^2 v^2 \int dx \frac{i}{(4\pi)^{d/2}}\Gamma(2-d/2)\left(\frac{1}{\Delta}\right)^{2-d/2}$

$= 18i\lambda^2 v^2 \frac{\Gamma(2-d/2)}{(4\pi)^2} + (\text{finite terms})$

And

these factors are explained in integration of very first diagrams in this exercise.

 $= \frac{1}{2}\,(N-1)(-2i\lambda v)^2 \int \frac{d^dk}{(2\pi)^d}\frac{i}{k^2}\frac{i}{(k+p)^2}$, this integration was solved whe calculating 

v - here since 3-point function

$= 2(N-1)\lambda^2 v^2\left[\frac{\Gamma(2-d/2)}{(4\pi)^2}\,i + (\text{finite terms})\right]$

$= 2i\lambda^2 v^2 (N-1)\frac{\Gamma(2-d/2)}{(4\pi)^2} + (\text{finite terms})$

the sum of these two divergences give $2i\lambda^2 v^2 \frac{\Gamma(2-d/2)}{(4\pi)^2}(N+8)$

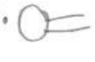
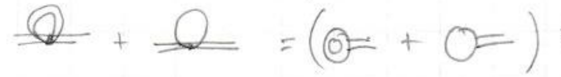
which is $-2i\,\delta_\lambda v^2$ which means the term $2i\delta_\lambda v^2$ of the counter term  cancels the divergencies of these two diagrams.

Let's now move on to the next two diagrams

 $= \frac{1}{2}\left(-6i\lambda\right)\int \frac{d^dk}{(2\pi)^d}\frac{i}{k^2-2|\mu|^2}$ which is exactly expression for $\frac{1}{v}\cdot$  of tadpole amplitude

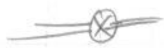
we will notice same for next diagram too

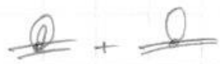
 $= \frac{1}{2}\left(-2i\lambda\delta^{ij}\right)\int \frac{d^dk}{(2\pi)^d}\frac{i\delta^{ij}}{k^2-\xi^2}$, where we put the infrared regulator ξ

We notice that this is same as the expression for $\frac{1}{v}\cdot$  so 

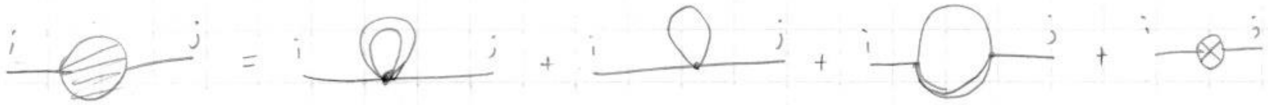
which has amplitude of $i(\delta_\mu - \delta_\lambda v^2)$ as we can see from our tadpole calculation

$\Rightarrow i(\delta_\mu - \delta_\lambda v^2) =$  which exactly cancels

the $-i(\delta_\mu - v^2\delta_\lambda)$ part of the  counter term

which means the divergencies of  are cancelled

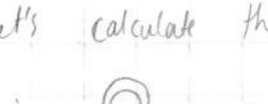
now we are left we no more divergent integrals, which means no divergent term proportional to p^2 $\Rightarrow$ $\delta_Z = 0$ at 1-loop order. We have now fixed all counter terms and show that divergencies cancel for all but one amplitude. Now let's show the cancellation for this last amplitude that is 2π amplitude:



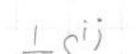
These are all of the 1-loop interactions since $(\pi^k)^3$ - terms aren't in Lagrangian. The counter term is $\delta^{ij}i\delta_Z p^2 - i(\delta_\mu - v^2\delta_\lambda)\delta^{ij}$ = i —⊗— j

i —⊗— j $= -i(\delta_\mu - v^2\delta_\lambda)\delta^{ij}$

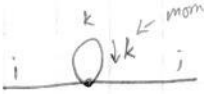
Let's calculate the integrals:

 $= \frac{1}{2}\left(-2i\lambda\delta^{ij}\right)\int \frac{d^dk}{(2\pi)^d}\frac{i}{k^2-2|\mu|^2}$, this is the integral

we calculated in tadpole and we can write this diagram in terms of tadpole to get correct expression:

 $= \frac{1}{2}\left(-2i\lambda\delta^{ij}\right)\int \frac{d^dk}{(2\pi)^d}\frac{i}{k^2-2|\mu|^2} = \frac{1}{3v}\delta^{ij}$  $= -\lambda i\,\frac{\Gamma(1-d/2)\;\delta^{ij}}{(2|\mu|^2)^{1-d/2}(4\pi)^{d/2}}$

Similarily

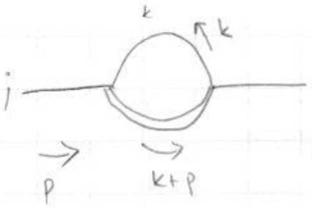
 $= \frac{1}{2}\left(-2i\lambda\left[\underbrace{\delta^{ij}\overset{=N-1}{\delta^{kk}}}_{=\delta^{ij}} + \delta^{ik}\delta^{jk} + \underbrace{\delta^{ik}\delta^{jk}}_{\delta^{ij}}\right]\right)\int\frac{d^dk}{(2\pi)^d}\frac{i}{k^2-\xi^2}$

$= \frac{1}{2}\left(-2i\lambda(N+1)\delta^{ij}\right)\int\frac{d^dk}{(2\pi)^d}\frac{i}{k^2-\xi^2} =$ ‖ let's use result for tadpole O= integral result or just use (A.44) of Peskin

$= -i\lambda(N+1)\,\frac{\Gamma(1-d/2)}{(\xi^2)^{1-d/2}}\,\frac{\delta^{ij}}{(4\pi)^{d/2}}$

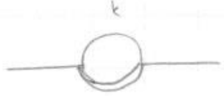
$$\Rightarrow \quad i \;\; j = -i\lambda (N+1)\,\delta^{ij}\,\frac{\Gamma(1-d/2)}{(4\pi)^{d/2}}\left(\frac{1}{\xi^2}\right)^{1-d/2}$$

Now let's look into the final loop diagram



$$= (-2i\lambda v\,\delta^{ik})(-2i\lambda v\,\delta^{jk})\int\frac{d^dk}{(2\pi)^d}\,\frac{i}{k^2-\xi^2}\,\frac{i}{(k+p)^2-2|\mu|^2}$$

The divergence of this diagram doesn't depend on p since it will be in $\Gamma(2-d/2)$ term after integration, and we are only interested in divergent part so let's set $p=0$



$$\Rightarrow \quad i \;\; j = 4\lambda^2 v^2\,\delta^{ij}\int\frac{d^dk}{(2\pi)^d}\,\frac{1}{k^2-\xi^2}\,\frac{1}{k^2-2|\mu|^2}$$

Now let's use feynman parametrization

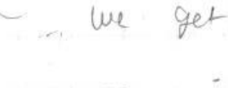
$$\text{loop} = 4\lambda^2 v^2\,\delta^{ij}\int dx\int\frac{d^dk}{(2\pi)^d}\,\frac{1}{D^2} \qquad \text{with} \qquad D = x(k^2-\xi^2)+(1-x)(k^2-2|\mu|^2)$$
$$= k^2 - x\xi^2 - 2(1-x)|\mu|^2$$

$$\Rightarrow \text{loop} = 4\lambda^2 v^2\,\delta^{ij}\int dx\int\frac{d^dk}{(2\pi)^d}\,\frac{1}{(k^2-\Delta)^2} \qquad \text{with} \quad \Delta = x\xi^2 + 2(1-x)|\mu|^2$$

Now using Minkowski integrals of (A.44) – Peskin we get

$$\text{loop} = 4\lambda^2 v^2\,\delta^{ij}\int dx\,\frac{i}{(4\pi)^{d/2}}\,\Gamma(2-d/2)\left(\frac{1}{\Delta}\right)^{2-d/2}$$
$$= 4\lambda^2 v^2\,\delta^{ij}\,\frac{\Gamma(2-d/2)}{(4\pi)^2} + \text{finite terms}$$

Summing the divergent parts of the diagrams along with the counter terms we get



$$i \;\; j \Bigg|^{\text{divergent}}_{p=0} = (-i\lambda\,\delta^{ij})\Bigg\{\frac{\Gamma(1-d/2)}{(4\pi)^{d/2}}\left[\frac{1}{(2|\mu|^2)^{1-d/2}}+\frac{N+1}{(\xi^2)^{1-d/2}}\right]$$
$$-4\lambda v^2\,\frac{\Gamma(2-d/2)}{(4\pi)^{d/2}} - \frac{\Gamma(1-d/2)}{(4\pi)^{d/2}}\left[\frac{3}{(2|\mu|^2)^{1-d/2}}+\frac{N-1}{(\xi^2)^{1-d/2}}\right]\Bigg\}$$
$$= (-i\lambda\,\delta^{ij})\Bigg\{2\frac{\Gamma(1-d/2)}{(4\pi)^{d/2}}\left[\frac{1}{(\xi^2)^{1-d/2}}-\frac{1}{(2|\mu|^2)^{1-d/2}}\right] - 4\lambda v^2\,\frac{\Gamma(2-d/2)}{(4\pi)^{d/2}}\Bigg\}$$

Now let's examine the behaviour of the term proportional to $\Gamma(1-d/2)$ at pole near $d=2$

$$2\,\frac{\Gamma(1-d/2)}{(4\pi)^{d/2}}\left[\frac{1}{(\xi^2)^{1-d/2}}-\frac{1}{(2|\mu|^2)^{1-d/2}}\right]$$

Taylor series $f^x \simeq 1-$

A simple pole times term that goes to 0 is always zero

Near $d=2$ the term inside the square brackets approach 0 as $d\to 2$ and cancel the simple pole in $\Gamma(1-d/2)$ so the divergence at $d=2$ cancels and we can expand $\Gamma(x)=\Gamma(x+1)/x$

$\Rightarrow$ $\Gamma(1-d/2)=\Gamma(2-d/2)/(1-d/2)$. and now the amplitude (with only divergent parts of diagrams) is

$$\text{Amp} = -i\lambda\delta^{ij}\left\{\frac{2\,\Gamma(2-d/2)}{(1-d/2)(4\pi)^{d/2}}\left[\frac{\xi^2}{(\xi^2)^{2-d/2}}-\frac{2|\mu|^2}{(2|\mu|^2)^{2-d/2}}\right]-4\lambda v^2\,\frac{\Gamma(2-d/2)}{(4\pi)^{d/2}}\right\}$$

$$= -2i\lambda\delta^{ij}\,\frac{\Gamma(2-d/2)}{(4\pi)^{d/2}}\left[\frac{2\xi^2}{(1-d/2)(\xi^2)^{2-d/2}}-\frac{4|\mu|^2}{(2|\mu|^2)^{2-d/2}}\cdot\frac{1}{1-d/2}-4\lambda v^2\right]$$

for $d>2$ and $\xi\to 0$ we can neglect first term in bracket
let's also use $v^2=\frac{|\mu|^2}{\lambda}$

$$\Rightarrow\quad \text{Amp} = -2i\lambda\delta^{ij}\,\frac{\Gamma(2-d/2)}{(4\pi)^{d/2}}\left[-\frac{4|\mu|^2}{(1-d/2)(2|\mu|^2)^{2-d/2}}-4|\mu|^2\right]$$

There is a simple pole at $d=4$ in $\Gamma(2-d/2)$ but at $d=4$ the terms in bracket give

$$-\frac{4|\mu|^2}{\underbrace{(1-2)\cdot 1}_{=-1}}-4|\mu|^2 = 4|\mu|^2-4|\mu|^2=0$$

So the terms in bracket go to zero at $d=4$ and the divergence vanishes since the pole is simple

$\Rightarrow$  is finite since divergence vanishes. (at least to 1-loop)

If we took into account the finite terms we could show the amplitude itself vanishes but we are only interested in the divergence vanishing so we let it be.

$\Rightarrow$ We have shown that with only the three counter-terms δ_Z, δ_μ and δ_λ all the divergent terms cancel (to 1-loop).